

Product Requirements Document (PRD)

AI-Native Restaurant Ecosystem v1.0

1. Executive Summary

This project aims to develop a next-generation restaurant website that moves beyond static menus. It leverages **Agentic AI** to handle customer service, complex ordering, and real-time engagement, providing a "high-tech, high-touch" experience for a global customer base.

2. Target Audience

- **Dine-in/Take-out Customers:** Individuals looking for a frictionless, fast ordering experience.
- **Global Travelers:** Users requiring multi-currency support and AI assistance for menu translations/ explanations.
- **Restaurant Staff:** Internal users needing automated order management and data insights.

3. Functional Requirements

3.1 AI Concierge (Chatbot)

- **Natural Language Ordering:** Users can type or speak orders (e.g., "Add two jollof rice but make one extra spicy").
- **Real-time Inventory Link:** Bot checks the database to ensure items are in stock before confirming.
- **Personalized Recommendations:** AI suggests pairings based on the current cart (e.g., "Would you like a drink with that?").
- **FAQ & Reservations:** Handles table bookings and questions about allergens or parking.

3.2 Digital Menu & Ordering Flow

- **Dynamic Menu:** Pulls live data from the database; supports "out of stock" toggles.
- **Cart Management:** Seamless addition/removal of items with real-time price updates.

- **Global Payments:** Integration with Stripe/Paystack for local and international transactions.

3.3 Admin & Data Insights

- **Order Management Dashboard:** Real-time view of incoming orders for kitchen staff.
- **Data Analytics:** Automated reports on top-selling items and peak hours (optimized for ETL pipelines).

4. Technical Specifications

Component	Requirement
Frontend	Next.js 14+ (React), Tailwind CSS, Mobile-First Design.
Backend	Node.js or Python (FastAPI) for high-performance AI orchestration.
Database	MSSQL or PostgreSQL (Relational structure for Menu, Users, and Orders).
AI Engine	Gemini 1.5 Pro or GPT-4o API via Function Calling.
Authentication	NextAuth.js (Support for Google, Email, and Guest Checkout).

5. User Experience (UX) Goals

"The goal is 'Zero Friction.' A user should be able to complete an order in under 60 seconds, whether via the AI bot or the manual menu."

- **Accessibility:** High contrast, screen-reader friendly, and voice-to-text enabled.
- **Performance:** Page load speed < 1.5 seconds; AI response time < 2 seconds.

6. Success Metrics (KPIs)

- **Conversion Rate:** Percentage of site visitors who complete an order.
- **AI Autonomy:** Percentage of orders completed via chatbot without human intervention.
- **Average Order Value (AOV):** Increase in AOV through AI-driven upselling.